A LOCALLY ALGEBRAIC GROUP SCHEME OVER A CHARACTERISTIC-ZERO FIELD IS SMOOTH

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Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-module-differentials (canonical tag 047I). cited at proof lines 1491-1493. The module of differentials Omega_{G/k} is free.
- Lemma varieties-lemma-char-zero-differentials-free-smooth (canonical tag 04QN). cited at proof lines 1494-1495. Characteristic zero plus free differentials imply smoothness.

Source reference. Original lemma 047I: sources/stacks-chow/tags.txt, source lines 5509–5509. Cited at proof lines 1491-1493: Omega_{G/k} is free; load-bearing.

Source reference. Original lemma 04QN: sources/stacks-chow/tags.txt, source lines 6156–6156. Cited at proof lines 1494-1495: free differentials plus characteristic zero give smoothness; load-bearing.

Lemma 0.2. *Let k be a field of characteristic 0. Let G be a locally algebraic group scheme over k. Then the structure morphism $G \to \mathrm{Spec}(k)$ is smooth, i.e., G is a smooth group scheme.*

Proof. By Lemma 047I the module of differentials of G over k is free. Hence smoothness follows from Varieties, Lemma 04QN. □

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